

TASK 2

Feature Engineering, Model Optimization & Performance Comparison

The task focuses on comparing different regression models for predicting house prices using the California Housing dataset. Two feature scaling techniques, StandardScaler and MinMaxScaler, were applied before training the models. The performance of Linear Regression, Ridge Regression, and Decision Tree Regressor was evaluated using quantitative performance metrics and visualization techniques. The study aims to identify the most suitable model for accurate house price prediction while understanding the effect of feature scaling on different machine learning algorithms.

Task Objective

- To load and preprocess the California Housing dataset.
- To apply feature scaling using StandardScaler and MinMaxScaler.
- To train multiple regression models for house price prediction.
- To compare model performance using RMSE and R^2 Score.
- To visualize model predictions using Actual vs Predicted scatter plots.
- To identify the best-performing regression model based on evaluation metrics.

Methodology

The California Housing dataset was imported using Scikit-learn. The dataset contains eight independent variables:

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

The target variable is **House Price**.

The dataset was divided into training data (80%) and testing data (20%) using the `train_test_split()` function with a random state of 42 to ensure reproducibility. After preprocessing, feature scaling was applied using two different scaling techniques. Three regression models were then trained separately using each scaling method.

Finally, predictions were generated on the testing dataset and evaluated using RMSE and R^2 Score. Scatter plots of Actual vs Predicted values were created to visually compare the prediction capability of each model.

Feature Scaling

Feature scaling is an important preprocessing step that transforms numerical features into comparable ranges, helping many machine learning algorithms perform more efficiently.

StandardScaler

StandardScaler standardizes each feature by removing its mean and scaling it to unit variance. It is suitable for algorithms that assume normally distributed data.

MinMaxScaler

MinMaxScaler transforms each feature into a fixed range between 0 and 1. It preserves the relative relationships among the data points while ensuring that all features have the same scale.

Both scaling techniques were applied in this project to compare their impact on model performance.

Models Used

1. Linear Regression

Linear Regression is a supervised learning algorithm that models the linear relationship between input variables and the target variable. It is simple, fast, and highly interpretable.

2. Ridge Regression

Ridge Regression is an extension of Linear Regression that applies L2 regularization. This helps reduce overfitting by penalizing large coefficient values, resulting in a more stable model.

3. Decision Tree Regressor

Decision Tree Regressor is a non-linear machine learning algorithm that recursively splits the dataset into smaller subsets based on feature values. It effectively captures complex and non-linear relationships within the data.

Performance Measures Used

Root Mean Square Error (RMSE)

RMSE measures the average prediction error between the actual and predicted values. Lower RMSE values indicate better model performance.

Coefficient of Determination (R^2 Score)

R^2 Score measures how well the model explains the variance in the target variable. Higher R^2 values indicate better prediction capability, with a maximum possible value of 1.

Interpretation of Visualization Charts

Type of Visualization Used

The prediction performance of each regression model was analysed using scatter plots of Actual House Prices vs Predicted House Prices. In each plot, the horizontal axis represents the actual house prices from the test dataset, while the vertical axis represents the prices predicted by the trained model.

A red diagonal line (45° reference line) was included in every plot to represent perfect predictions, where the predicted value is exactly equal to the actual value.

Purpose of the Visualization

The scatter plots were generated to visually evaluate and compare the prediction accuracy of different machine learning models. These plots help to:

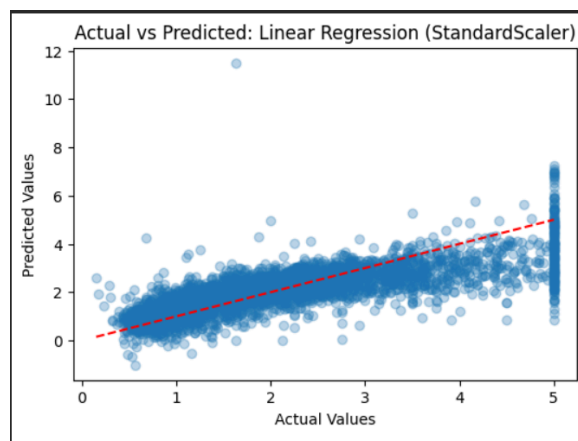
- Compare the closeness of predicted values to the actual values.

- Identify prediction errors and deviations from the ideal prediction line.
- Examine whether a model captures the relationship between the features and house prices effectively.
- Compare the influence of different feature scaling techniques on model performance.
- Detect under-prediction, over-prediction and non-linear prediction behaviour.

A model with prediction points clustered closely around the red reference line is considered to have better prediction accuracy.

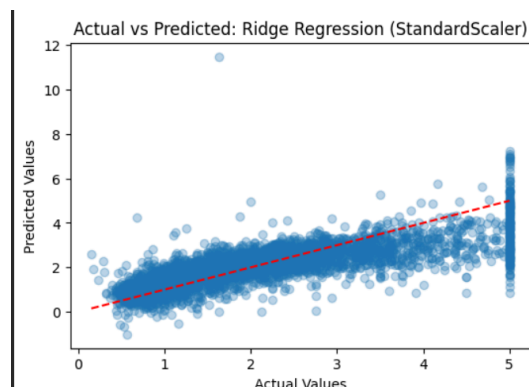
Interpretation of Individual Visualizations

1. Linear Regression with StandardScaler



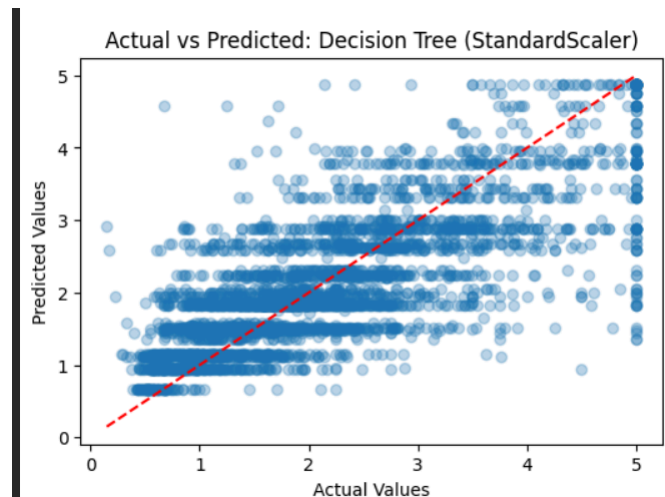
The scatter plot shows that most prediction points follow the reference line, indicating that the model successfully captures the overall linear relationship between the housing features and house prices. However, the points become more scattered for higher-priced houses, suggesting that the model struggles to accurately predict extreme values. The moderate spread around the reference line indicates reasonable prediction accuracy but also highlights the limitations of using a purely linear model for a dataset containing more complex relationships.

2. Ridge Regression with StandardScaler



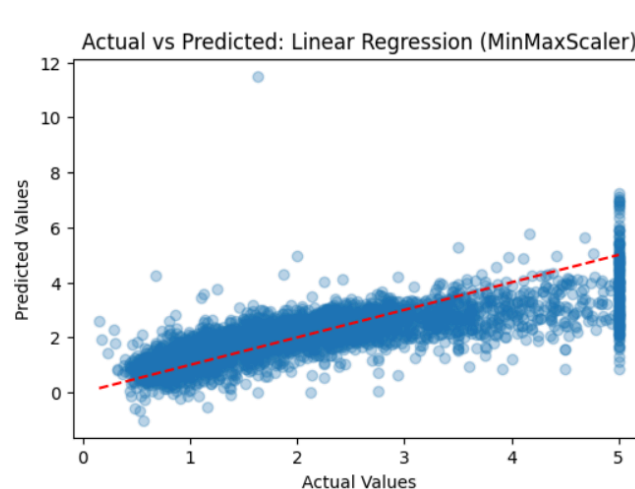
The visualization is very similar to that of Linear Regression because Ridge Regression is an extension of the linear model with L2 regularization. The prediction points remain close to the reference line, while regularization helps improve model stability by reducing the effect of large coefficients. Although the overall prediction pattern is consistent, the improvement over Linear Regression is minimal, indicating that regularization alone does not significantly enhance prediction performance for this dataset.

3. Decision Tree Regressor with StandardScaler



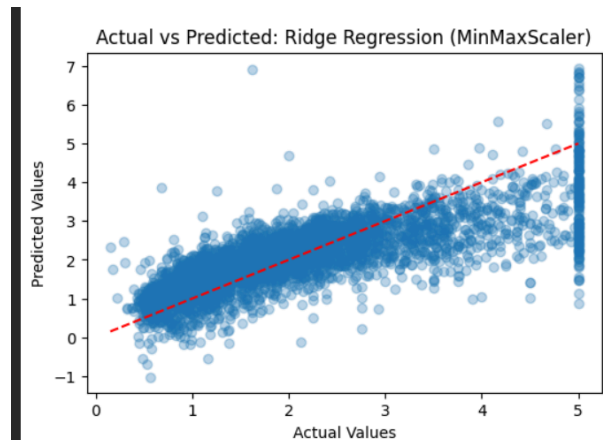
Unlike the linear models, the Decision Tree Regressor produces prediction points arranged in horizontal bands. This occurs because decision trees predict constant values for all observations within the same leaf node. Despite this characteristic, the prediction points are generally closer to the reference line across different price ranges, showing that the model captures non-linear relationships more effectively. The smaller deviations from the reference line indicate better prediction accuracy than the linear models.

4. Linear Regression with MinMaxScaler



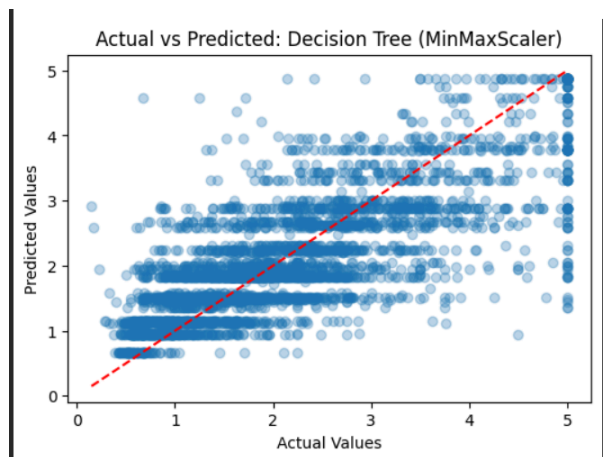
The scatter plot closely resembles the StandardScaler version, with prediction points distributed similarly around the reference line. This indicates that changing the scaling technique has almost no influence on the performance of Linear Regression. The model continues to capture the overall trend but exhibits noticeable prediction errors for houses with very high prices.

5. Ridge Regression with MinMaxScaler



The prediction pattern is slightly improved compared with Ridge Regression using StandardScaler. More prediction points appear closer to the reference line, indicating a small improvement in prediction accuracy. This observation is also supported by the reduction in RMSE and the increase in R^2 Score. The model remains stable while benefiting modestly from MinMax feature scaling.

6. Decision Tree Regressor with MinMaxScaler



The visualization is almost identical to the Decision Tree model trained using StandardScaler. The same horizontal prediction bands are observed, and the prediction points remain close to the reference line. This confirms that Decision Tree algorithms are largely insensitive to feature scaling, producing nearly identical predictions regardless of the scaling technique applied.

- Linear Regression produced consistent prediction patterns with both StandardScaler and MinMaxScaler. The predictions generally followed the ideal reference line but showed increasing errors for higher-priced houses, indicating that the model was limited in capturing complex, non-linear relationships.
- Ridge Regression demonstrated prediction behaviour very similar to Linear Regression. However, when MinMaxScaler was applied, the prediction points appeared slightly closer to the reference line, suggesting a modest improvement in model performance due to feature scaling combined with regularization.
- Decision Tree Regressor produced distinctly different scatter plots characterized by horizontal prediction bands, which are expected because tree-based models generate piecewise constant predictions. Despite this appearance, the prediction points remained closer to the reference line than those of the linear models, indicating superior predictive performance.
- Comparing the two feature scaling techniques shows that StandardScaler and MinMaxScaler had little effect on Decision Tree Regressor, whereas MinMaxScaler provided a slight improvement for Ridge Regression. Linear Regression was largely unaffected by the choice of scaling technique.

Overall, the visualization analysis clearly indicates that the Decision Tree Regressor generated predictions that were closest to the actual house prices among all six model–scaler combinations.

Comparison Table

Combined Model Evaluation Results:				
	Scaler	Model	RMSE	R ²
0	StandardScaler	Linear Regression	0.745581	0.575788
1	StandardScaler	Ridge Regression	0.745557	0.575816
2	StandardScaler	Decision Tree	0.705184	0.620513
3	MinMaxScaler	Linear Regression	0.745581	0.575788
4	MinMaxScaler	Ridge Regression	0.738520	0.583785
5	MinMaxScaler	Decision Tree	0.705184	0.620513

The comparison table shows that the Decision Tree Regressor achieved the best overall performance under both StandardScaler and MinMaxScaler, obtaining the lowest RMSE (0.705184) and the highest R² Score (0.620513). This indicates that the model predicted house prices more accurately than the other regression models.

Linear Regression produced identical results with both scaling techniques, demonstrating that feature scaling had little influence on its performance. Ridge Regression showed a slight

improvement when MinMaxScaler was applied, reducing the RMSE from 0.745557 to 0.738520 and increasing the R^2 Score from 0.575816 to 0.583785.

The visualization charts support these numerical findings. Linear Regression and Ridge Regression captured the general linear relationship within the dataset but showed larger prediction errors for higher house prices. In contrast, the Decision Tree Regressor better captured non-linear patterns and generated predictions that were closer to the actual values.

Conclusion

This project successfully implemented feature engineering, feature scaling, model training, performance evaluation, and visualization for house price prediction using the California Housing dataset.

Among the three regression models evaluated, the Decision Tree Regressor consistently achieved the best performance with both StandardScaler and MinMaxScaler, recording the lowest RMSE (0.705184) and the highest R^2 Score (0.620513). The model effectively captured the complex, non-linear relationships within the dataset, resulting in more accurate predictions than the linear models.

Linear Regression provided a simple and interpretable baseline model with consistent performance across both scaling methods. Ridge Regression benefited slightly from MinMaxScaler due to regularization, but its overall accuracy remained lower than that of the Decision Tree.

Overall, this study demonstrates that selecting an appropriate machine learning algorithm has a greater impact on prediction accuracy than the choice of feature scaling technique for this dataset. The Decision Tree Regressor is therefore recommended as the most suitable model for this house price prediction task. Future improvements may include hyperparameter tuning and the evaluation of advanced ensemble models such as Random Forest, Gradient Boosting, and XGBoost to further enhance predictive performance.